Pseudocode

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PROGRAM TicTacToe

    // --- MAIN PROGRAM ---
    START Main
        DISPLAY Header "TIC TAC TOE PROJECT"

        READ n1 ("Player 1 Name")
        IF length of n1 > 10 THEN n1 = first 10 characters of n1
        IF n1 is empty THEN n1 = "Player 1"

        READ n2 ("Player 2 Name")
        IF length of n2 > 10 THEN n2 = first 10 characters of n2
        IF n2 is empty THEN n2 = "Player 2"

        SET scoreA = 0
        SET scoreB = 0
        SET keepPlaying = TRUE

        WHILE keepPlaying IS TRUE DO
            CALL ResetBoard()
            SET moves = 0
            SET gameOver = FALSE
            SET currentPlayer = 1  // 1 = Player 1 ('X'), 2 = Player 2 ('O')

            WHILE moves < 9 AND gameOver IS FALSE DO
                CALL RenderUI()

                IF currentPlayer IS 1 THEN
                    SET activeName = n1
                    SET symbol = 'X'
                ELSE
                    SET activeName = n2
                    SET symbol = 'O'
                END IF

                PROMPT activeName to enter position (1-9)
                READ inputKey

                IF inputKey is between '1' and '9' THEN
                    SET index = numerical value of inputKey - 1

                    IF board[index] != 'X' AND board[index] != 'O' THEN
                        board[index] = symbol
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            INCREMENT moves by 1

            IF CALL CheckWin(symbol) IS TRUE THEN
               SET gameOver = TRUE
               IF currentPlayer IS 1 THEN
                  INCREMENT scoreA by 1
               ELSE
                  INCREMENT scoreB by 1
               END IF
               CALL RenderUI()
               DISPLAY activeName + " Wins!"
            ELSE IF moves == 9 THEN
               SET gameOver = TRUE
               CALL RenderUI()
               DISPLAY "Draw Round!"
            ELSE
               // Switch player turn
               IF currentPlayer IS 1 THEN
                  currentPlayer = 2
               ELSE
                  currentPlayer = 1
               END IF
            END IF
         ELSE
            DISPLAY "Slot taken! Press any key..."
            WAIT for key press
         END IF
      ELSE
         DISPLAY "Invalid choice! Press any key..."
         WAIT for key press
      END IF
   END WHILE

   // Post-Game Menu
   SET choiceMade = FALSE
   WHILE choiceMade IS FALSE DO
      PROMPT "[1] End Game | [2] Restart Game: "
      READ choiceKey

      IF choiceKey == '1' THEN
         keepPlaying = FALSE
         choiceMade = TRUE
         CALL ShowFinalScores()
      ELSE IF choiceKey == '2' THEN
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            choiceMade = TRUE
            DISPLAY "Game Restarted..."
          ELSE
            DISPLAY "Invalid choice. Press 1 or 2."
          END IF
        END WHILE
    END WHILE
END Main


// --- SUBROUTINES / FUNCTIONS ---

FUNCTION ResetBoard
    FOR i FROM 0 TO 8 DO
        board[i] = character representation of (i + 1)
    END FOR
END FUNCTION


FUNCTION RenderUI
    CLEAR Screen
    DISPLAY Header "ITEC 102: TIC-TAC-TOE"
    DISPLAY Board Grid with board[0] through board[8]
    DISPLAY Player Names (n1, n2) and Current Scores (scoreA, scoreB)
END FUNCTION


FUNCTION CheckWin(mark)
    DEFINE winPatterns Matrix:
        [0, 1, 2], [3, 4, 5], [6, 7, 8],  // Top, Middle, Bottom Rows
        [0, 3, 6], [1, 4, 7], [2, 5, 8],  // Left, Center, Right Columns
        [0, 4, 8], [2, 4, 6]              // Diagonals

    FOR EACH pattern IN winPatterns DO
        IF board[pattern[0]] == mark AND
           board[pattern[1]] == mark AND
           board[pattern[2]] == mark THEN
            RETURN TRUE
        END IF
    END FOR

    RETURN FALSE
END FUNCTION
```

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    FUNCTION ShowFinalScores
        CLEAR Screen
        DISPLAY Header "FINAL SCORES"
        DISPLAY n1 + " Score: " + scoreA
        DISPLAY n2 + " Score: " + scoreB
        DISPLAY "Thank you for playing!"
    END FUNCTION

END PROGRAM
```